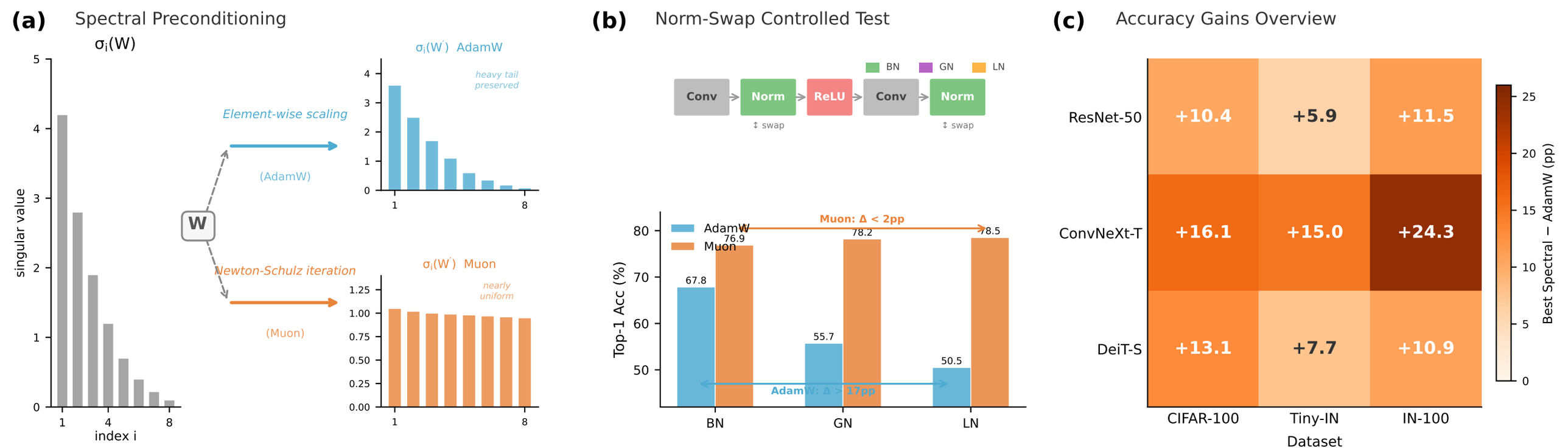
(a)
Spectral Preconditioning
σi(W)
singular value
index i
W
Element-wise scaling
(AdamW)
Newton-Schulz iteration
(Muon)
σi(W′) AdamW
heavy tail preserved
σi(W′) Muon
nearly uniform
(b)
Norm-Swap Controlled Test
Conv
Norm
ReLU
Conv
Norm
BN
GN
LN
↕ swap
↕ swap
Top-1 Acc (%)
AdamW
Muon
Muon: Δ < 2pp
AdamW: Δ > 17pp
67.8
76.9
55.7
78.2
50.5
78.5
BN
GN
LN
(c)
Accuracy Gains Overview
ResNet-50
ConvNeXt-T
DeiT-S
+10.4
+5.9
+11.5
+16.1
+15.0
+24.3
+13.1
+7.7
+10.9
CIFAR-100
Tiny-IN
IN-100
Dataset
Best Spectral − AdamW (pp)